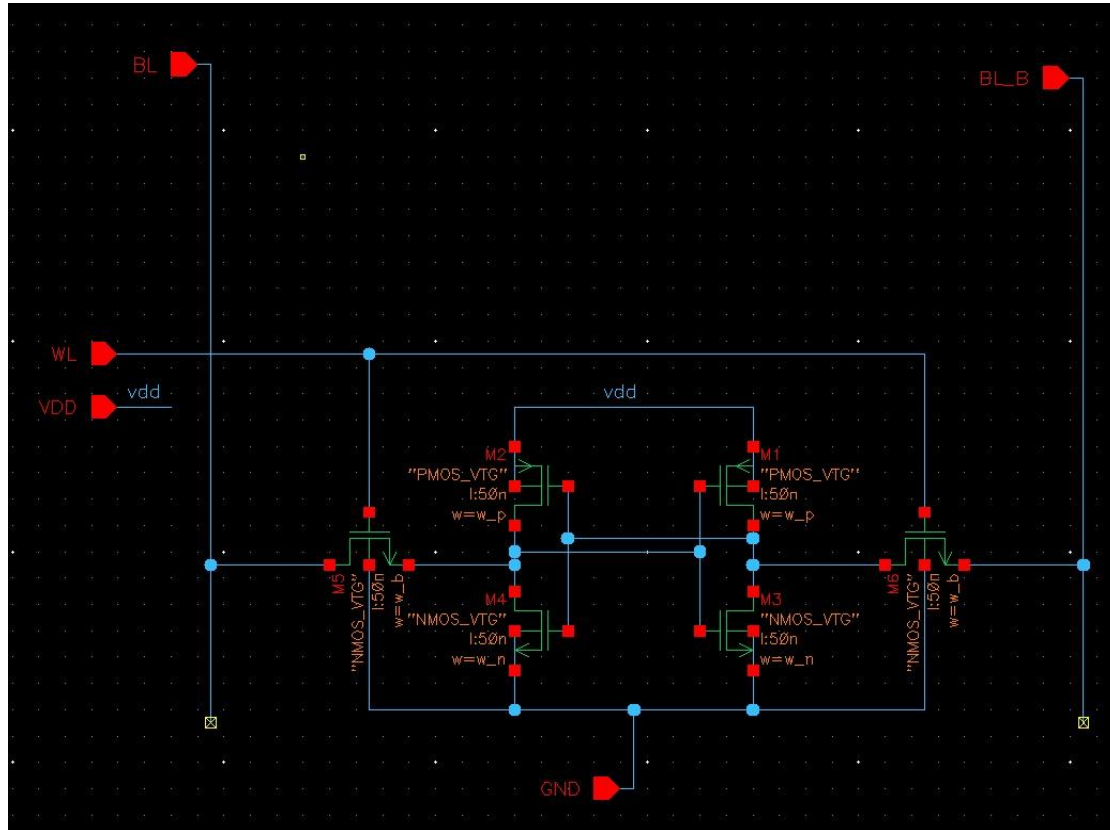
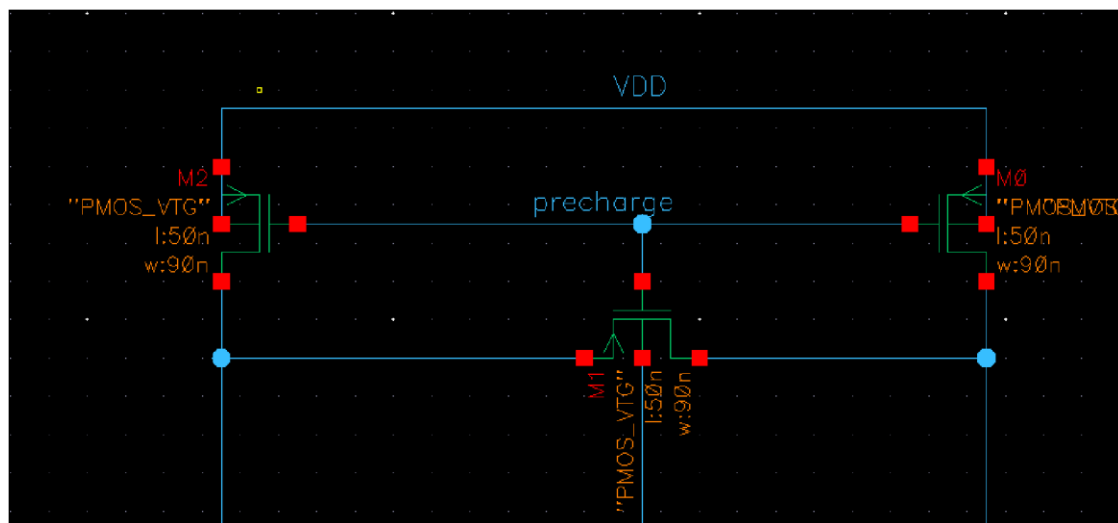


CE 391 Lab 3 Report

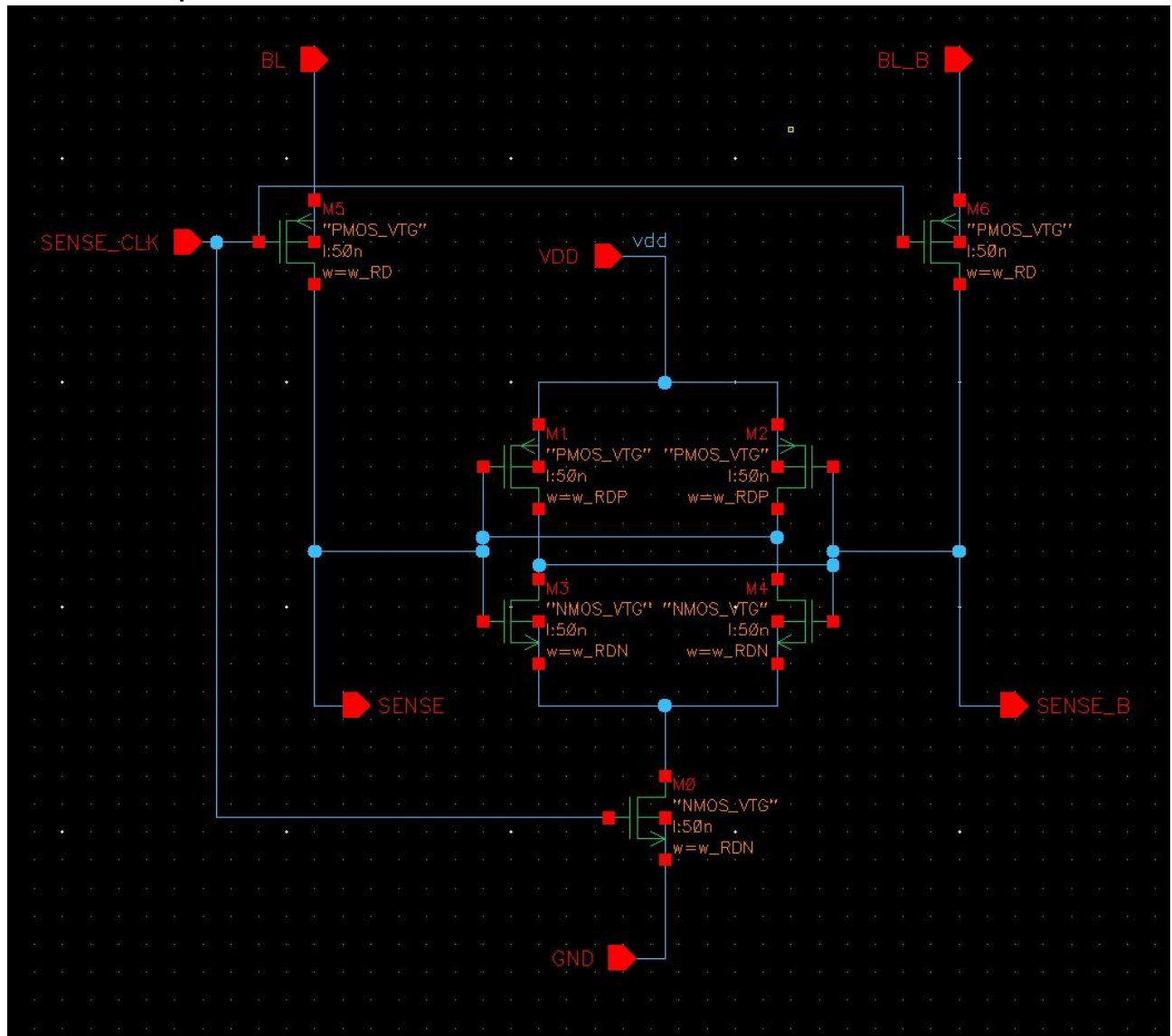
Schematics - SRAM Cell



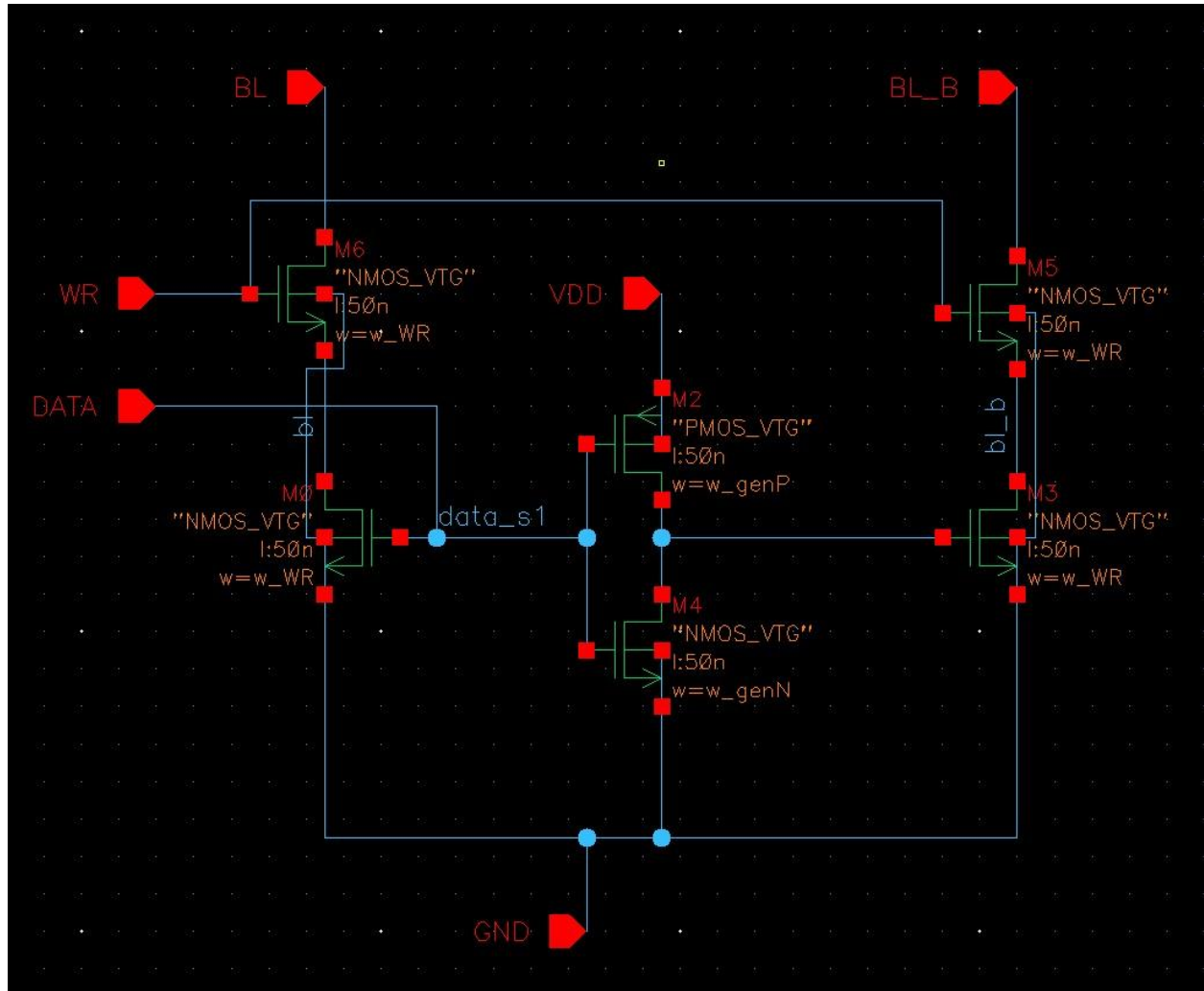
Bitline Conditioning



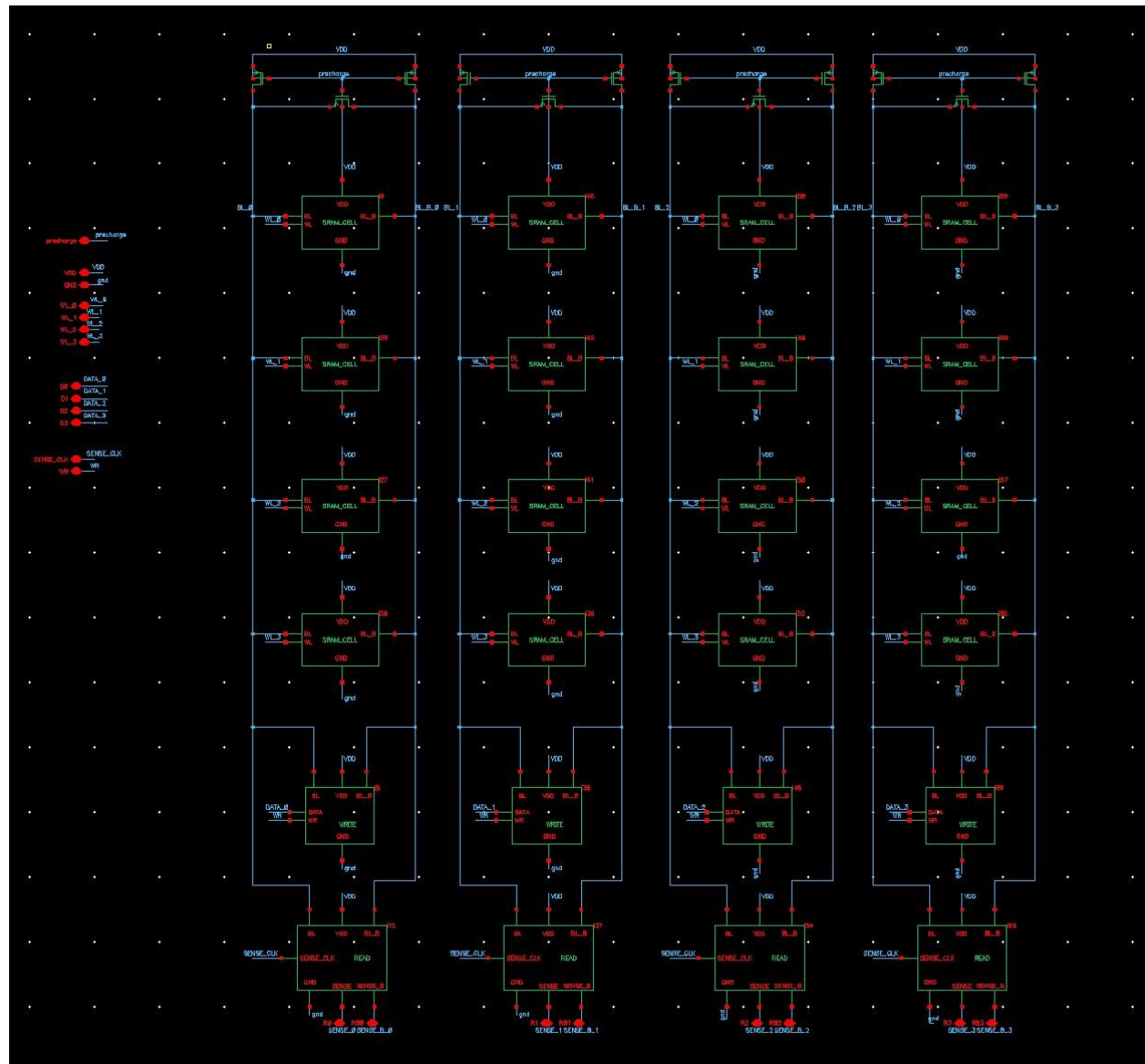
Sense Amplifier



Write Circuitry



4x4 6T SRAM Array with Sense Amplifier, bitline precharge and write on each column

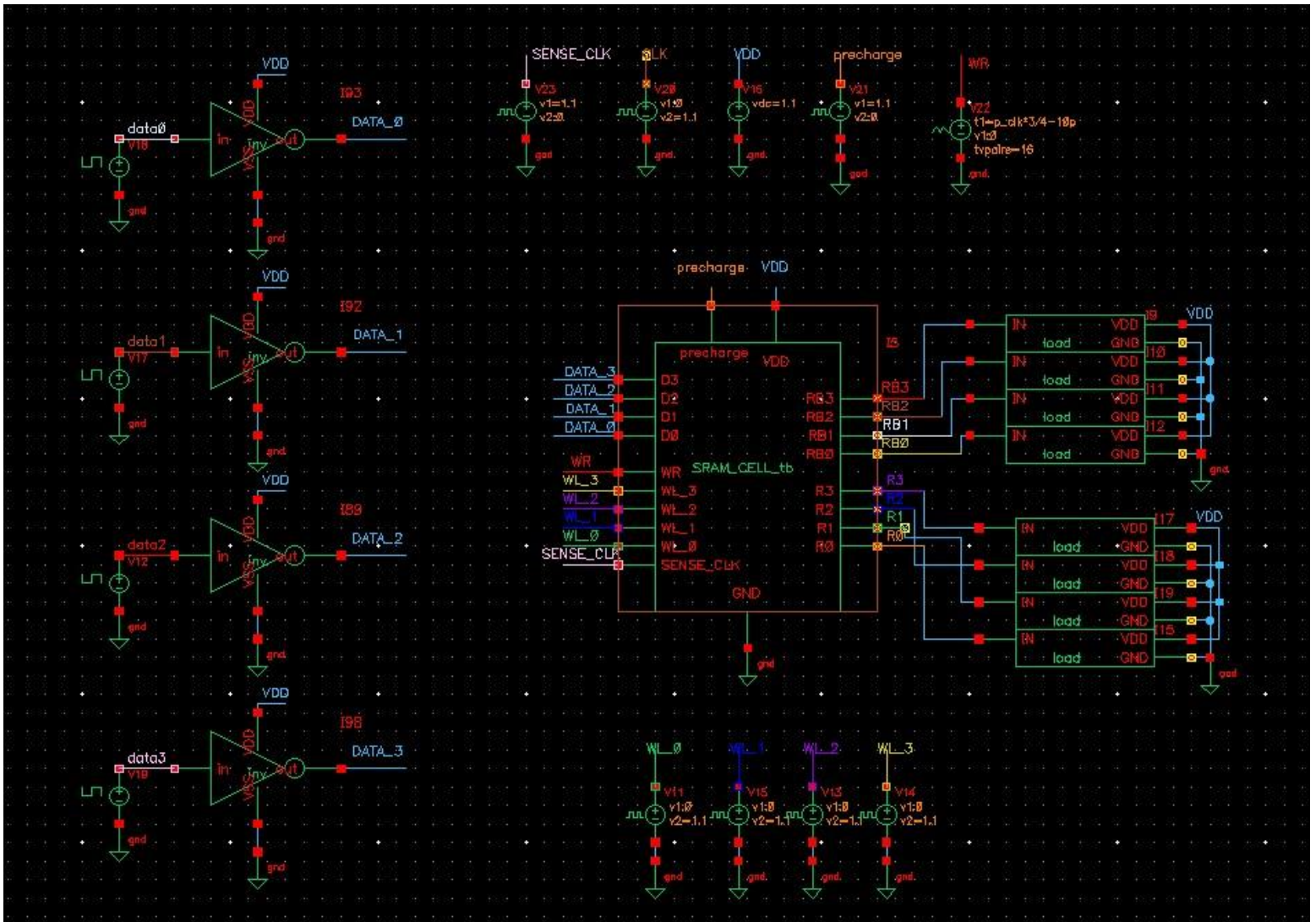


Functioning Test Bench

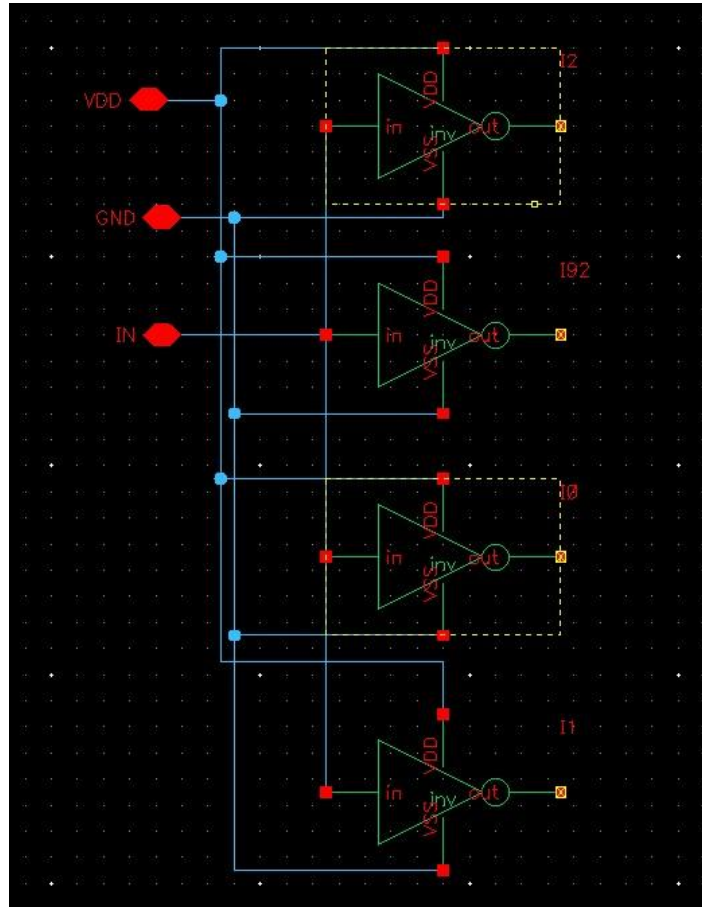
with logical write inputs of 0110, 1000, 0101, 1001 to each row. As you can see, each time the SENSECLK is pulsed high, the sense amp cuts off itself from the bitlines and is able to achieve a logical 0 or 1 depending on the value written into that bit.

Bit (Red) Bit_B(Green)





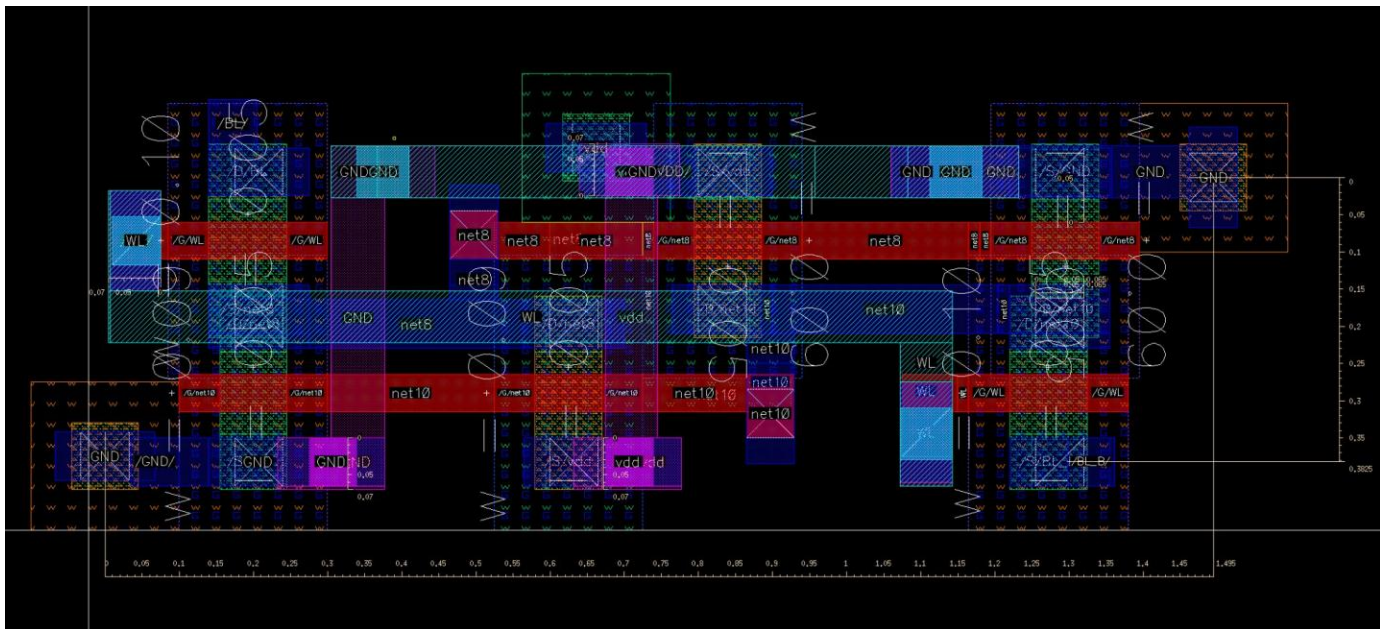
Four Inverter Load Symbol abstraction for ease of viewing the testbench



Layout, DRC, LVS

6T SRAM with dimensions 1.495um x 0.3825um = 0.5718375um²

I referenced a special area minimizing model presented by this paper [\[link\]](#)



Calibre - RVE v2011.4_35.27 : svdb SRAM_CELL

File View Highlight Tools Window Setup Help

Find: []

Navigator

Results

- Extraction Results
- Comparison Results

Reports

- Rules File
- Extraction Report
- LVS Report

View

- Info
- Finder
- Schematics

Setup

- Options

Comparison Results

Layout Cell / Type	Source Cell	Nets	Instances	Ports
SRAM_CELL	SRAM_CELL	SL, SS	1L, 1R	SL, SS

Cell SRAM_CELL Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

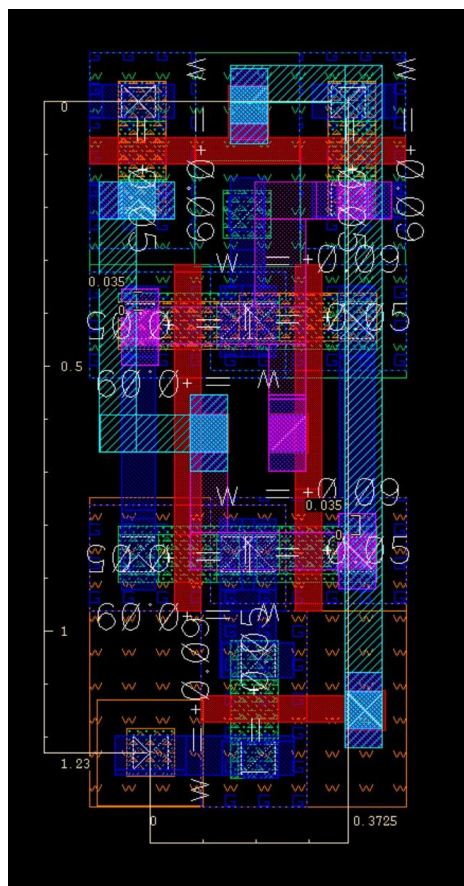
Layout Cell Name: SRAM_CELL
Source Cell Name: SRAM_CELL

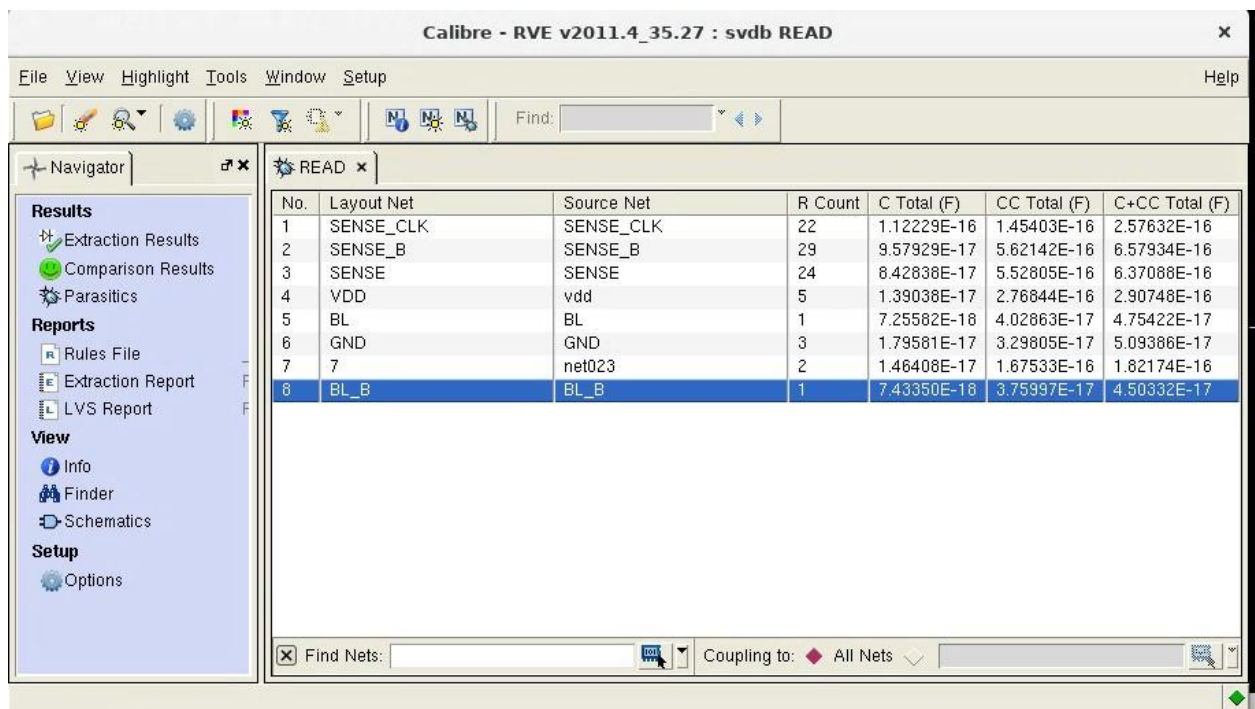
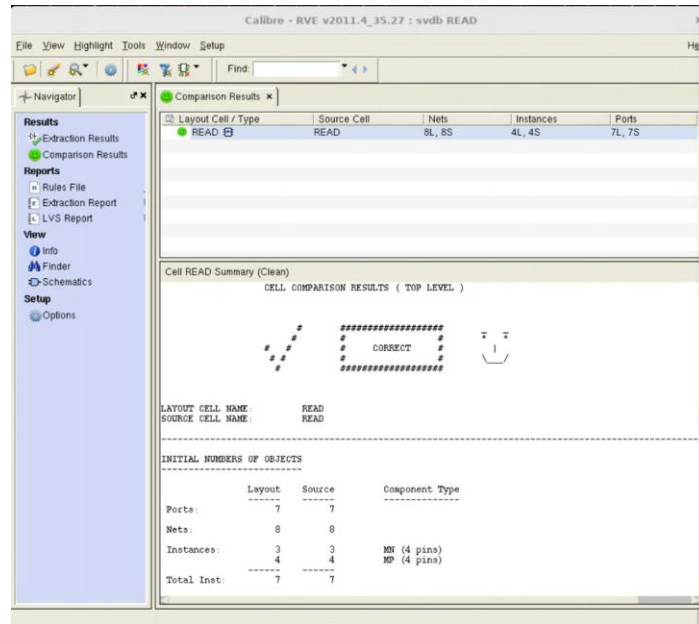
INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	5	5	
Nets:	7	7	
Instances:	4	4	1R (4 pins)
Total Inst:	6	6	1R (4 pins)

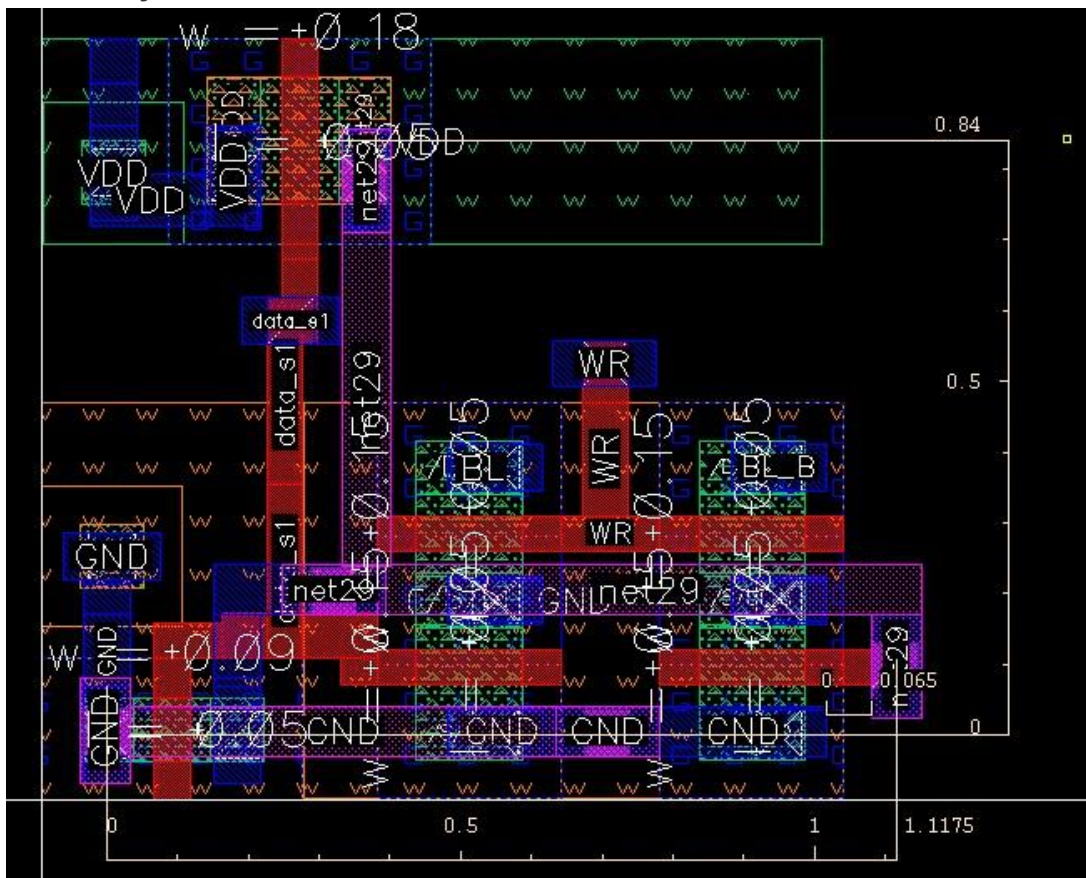
NUMBERS OF OBJECTS AFTER TRANSFORMATION

	Layout	Source	Component Type
Ports:	5	5	
Nets:	5	5	





Write Circuit Layout & DRC



Calibre - RVE v2011.4_35.27 : svdb WRITE

File View Highlight Tools Window Setup Help

Find

Navigator

Results

- Extraction Results
- Comparison Results

Reports

- Rules File
- Extraction Report
- LVS Report

View

- Info
- Finder
- Schematics

Setup

- Options

Comparison Results

Layout Cell / Type	Source Cell	Nets	Instances	Ports
WRITE	WRITE	7L_75	3L_35	6L_65

Cell WRITE Summary (Clean)

CELL COMPARISON RESULTS (TOP LEVEL)

LAYOUT CELL NAME: WRITE
SOURCE CELL NAME: WRITE

INITIAL NUMBERS OF OBJECTS

	Layout	Source	Component Type
Ports:	6	6	
Nets:	9	9	
Instances:	5	5	MP (4 pins)
	1	1	MP (4 pins)
Total Inst:	6	6	

Calibre - RVE v2011.4_35.27 : svdb Precharge

FileViewHighlightToolsWindowSetupHelp

Find:

Navigator

Results

Extraction Results

Comparison Results

Parasitics

Reports

Rules File

Extraction Report

LVS Report

View

Info

Finder

Schematics

Setup

Options

Precharge

No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	VDD	VDD	13	7.27803E-17	2.21782E-16	2.94563E-16
2	BL	BL	6	4.53292E-17	1.64791E-16	2.10120E-16
3	precharge	precharge	13	1.01872E-16	5.97864E-17	1.61658E-16
4	BL_B	BL_B	6	5.02074E-17	1.02756E-16	1.52963E-16

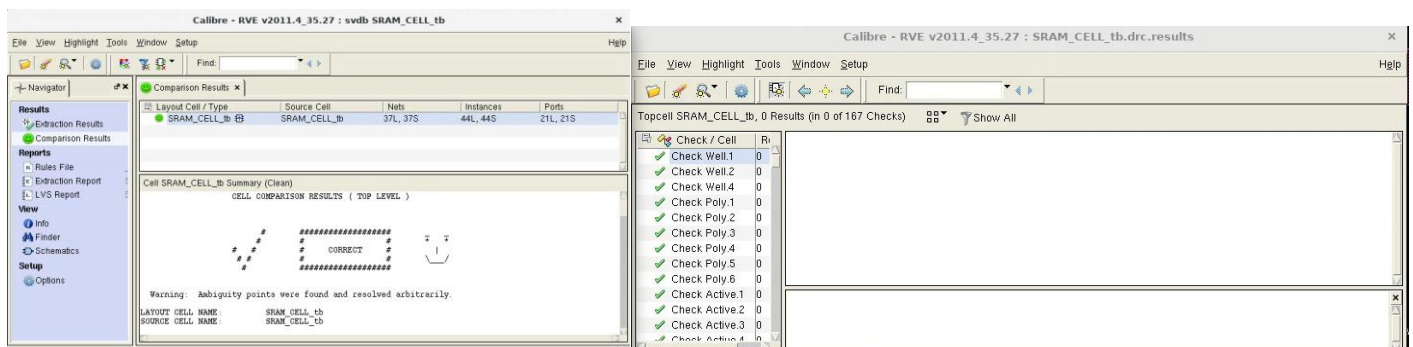
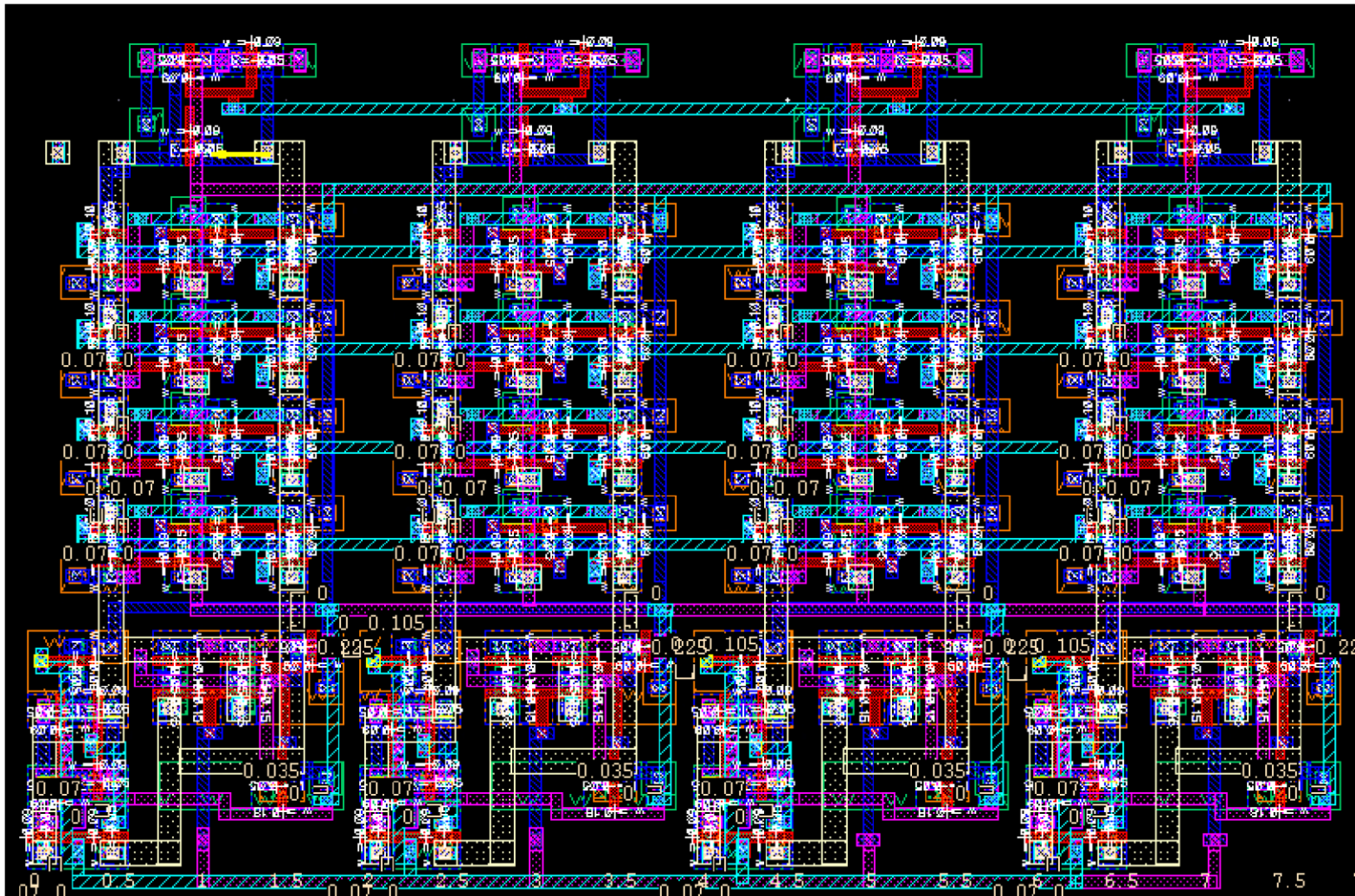
Find Nets:

Coupling to:

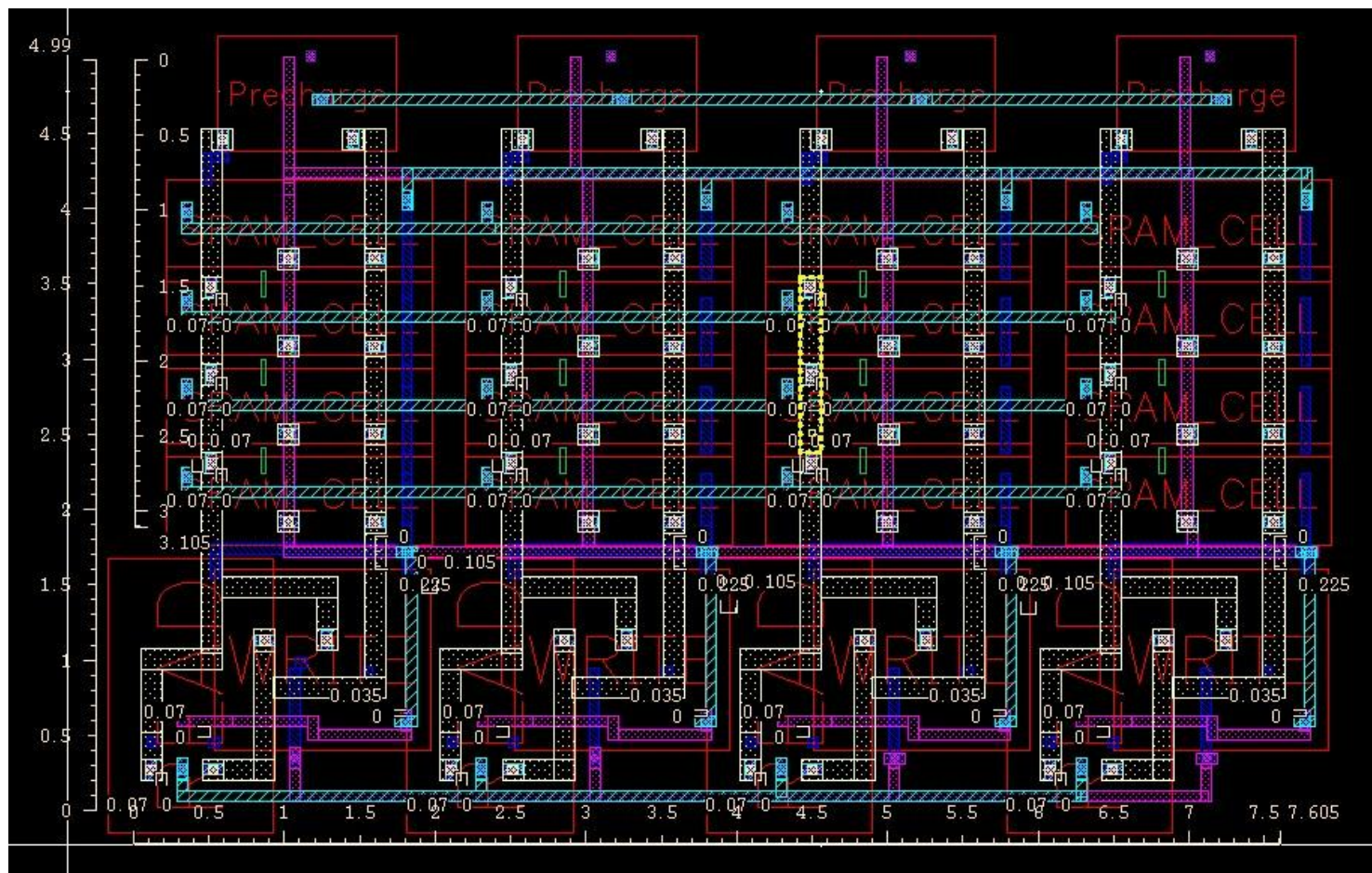
All Nets

4x4 cell

Total dimensions of 16 cells $2.4\mu\text{m} \times 7.6\mu\text{m} = 18.24\mu\text{m}^2$



With added Read, write and precharge: $4.99\mu\text{m} \times 7.6\mu\text{m}$



Calibre - RVE v2011.4_35.27 : svdb SRAM_CELL_tb

File View Highlight Tools Window Setup Help

Find:

Navigator

Results

- Extraction Results
- Comparison Results
- Parasitics

Reports

- Rules File
- Extraction Report
- LVS Report

View

- Info
- Finder
- Schematics

Setup

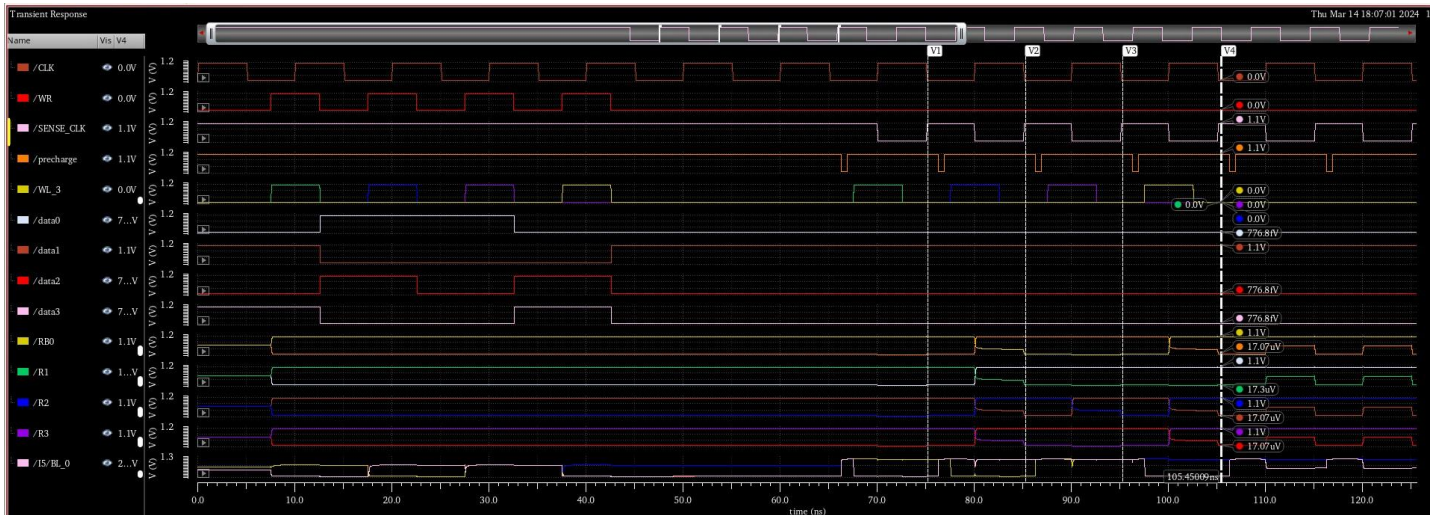
- Options

SRAM_CELL_tb

No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	1	BL_0	43	2.47001E-16	1.45249E-15	1.69949E-15
2	SENSE_CLK	SENSE_CLK	103	5.99442E-16	1.35010E-15	1.94954E-15
3	WL_3	WL_3	84	3.39436E-16	1.52969E-15	1.86913E-15
4	WL_2	WL_2	83	3.35232E-16	1.54073E-15	1.87596E-15
5	WL_1	WL_1	83	3.35688E-16	1.57379E-15	1.90948E-15
6	WL_0	WL_0	83	3.35171E-16	1.38314E-15	1.71831E-15
7	8	BL_B_0	44	2.36384E-16	1.44788E-15	1.68426E-15
8	VDD	VDD	429	1.34272E-15	6.82654E-15	8.16926E-15
9	WR	WR	65	4.95041E-16	6.42130E-16	1.13717E-15
10	precharge	precharge	59	5.85507E-16	5.72347E-16	1.15785E-15
11	GND	gnd	433	1.63753E-15	8.69772E-15	1.03352E-14
12	13	BL_1	43	2.26164E-16	1.57575E-15	1.80192E-15
13	14	BL_B_1	44	2.35285E-16	1.45431E-15	1.68960E-15
14	15	BL_2	43	2.21886E-16	1.53344E-15	1.75533E-15
15	16	BL_B_2	44	2.34906E-16	1.43157E-15	1.66647E-15
16	17	BL_3	43	2.21796E-16	1.53327E-15	1.75507E-15
17	18	BL_B_3	44	2.46104E-16	1.35216E-15	1.59827E-15
18	R0	SENSE_0	24	7.03742E-17	7.31658E-16	8.02032E-16
19	RB0	SENSE_B_0	27	7.82100E-17	7.24589E-16	8.02799E-16
20	D0	DATA_0	10	1.05855E-16	9.34631E-17	1.99318E-16
21	R1	SENSE_1	23	6.89561E-17	7.40293E-16	8.09249E-16
22	RB1	SENSE_B_1	27	7.63068E-17	7.38013E-16	8.14320E-16
23	D1	DATA_1	10	1.05855E-16	9.34631E-17	1.99318E-16
24	R2	SENSE_2	23	6.89561E-17	7.40293E-16	8.09249E-16
25	RB2	SENSE_B_2	27	7.63068E-17	7.38013E-16	8.14320E-16
26	D2	DATA_2	10	1.05855E-16	9.34631E-17	1.99318E-16
27	R3	SENSE_3	24	6.92501E-17	7.39709E-16	8.08959E-16
28	RB3	SENSE_B_3	27	7.65885E-17	7.37734E-16	8.14322E-16
29	D3	DATA_3	10	1.05956E-16	9.32915E-17	1.99247E-16
30	X0/7	XI108/net29	16	6.96525E-17	4.75595E-16	5.45248E-16
31	X0/8	XI108/net014	0	4.59044E-18	1.07371E-16	1.11961E-16
32	X0/9	XI108/net015	0	4.38959E-18	2.10128E-16	2.14517E-16
33	X1/7	XI109/net29	16	6.92851E-17	4.76735E-16	5.46020E-16
34	X1/8	XI109/net014	0	4.41410E-18	1.06836E-16	1.11250E-16
35	X1/9	XI109/net015	0	4.38959E-18	2.10128E-16	2.14517E-16
36	X2/7	XI110/net29	16	6.92851E-17	4.76479E-16	5.45764E-16
37	X2/8	XI110/net014	0	4.41410E-18	1.06836E-16	1.11250E-16
38	X2/9	XI110/net015	0	4.38959E-18	2.10128E-16	2.14517E-16
39	X3/7	XI111/net29	16	6.90738E-17	4.78397E-16	5.47470E-16

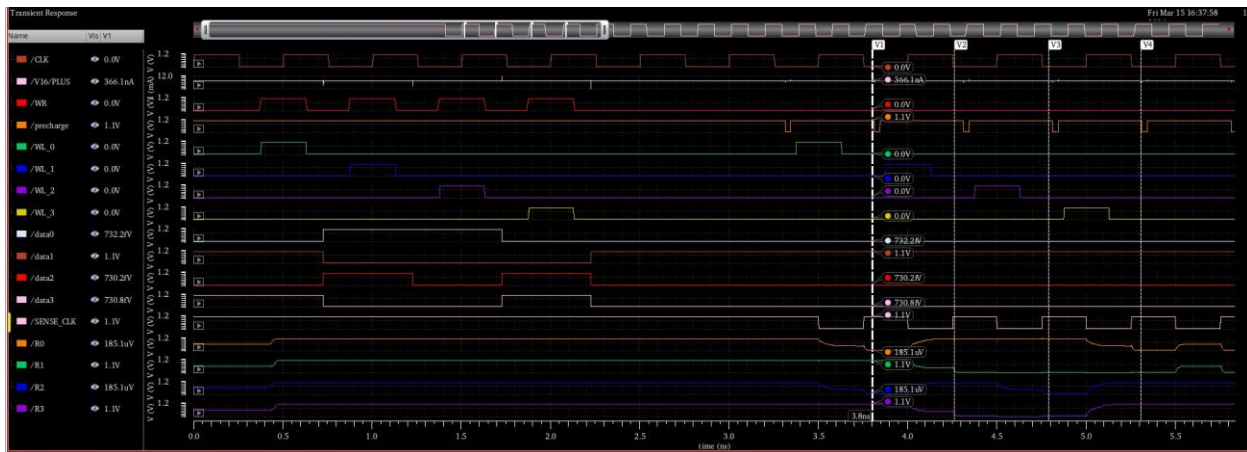
Find Nets: Coupling to: ◆ All Nets

Extracted simulation



Timing Limits

I ran majority of these simulations at the fastest achievable clk period of 10ns. The lowest functional clk period I tested was 0.5ns, as shown below. At lower periods, I observe the bitline transitions being too slow to accommodate read and write operations. Extracted circuit was able to run at a minimum of 5ns clk period.



I believe the time limiting factor here is probably the Sense Amp. With small transistor sizes in the SA, it is difficult for it to pull down super fast to sample the bitline signal. Normally with a much larger bit array, the bitlines would be the time limiting factor as it would have large parasitics, resulting in a large time constant when being pulled up or down.

Energy consumption: 34.31uW (schematic) and 45uW (layout) for the entire writing and reading of the 16bit sram. Divided down, for one single bit read/write it consumes 2.14uW and 2.81uW.